

KUET Kinship

Software Documentation — GitHub README Edition

Project Overview

KUET Kinship is an Android application designed to help KUET students access academic and campus-related information from one place. The current application focuses on course information, examination information, lecture notes, and rooms/labs with navigation support.

Project Goals

- Provide students with organized academic information.
- Make course and examination information easier to access.
- Provide a structured place for lecture notes.
- Help students find rooms and laboratories more easily.
- Keep the application simple and convenient for everyday student use.

Current Feature Scope

Feature	Current Status	Description
Course Selection	Implemented	Year → Term → Course selection and course information.
Examination	Implemented	Exam-related information and details.
Lecture Notes	Implemented	Access to available lecture notes.
Rooms & Labs	Implemented	Room/lab information with navigation support where available.
Updates	Not implemented	Reserved for future development.
Settings	Not implemented	Reserved for future development.

High-Level Architecture

- MainActivity acts as the main application entry point.
- HomeFragment provides the primary dashboard/navigation area.
- Academic flow: Course Selection → Year Selection → Term Selection → Course information.
- Examination flow: Exam section → relevant examination information → details.
- Learning flow: Lecture Notes → relevant academic/course information → lecture note.
- Campus flow: Rooms & Labs → room/lab information → navigation when available.

Main Components

Component	Responsibility
MainActivity	Initializes the main application structure and navigation.

HomeFragment	Displays the main dashboard and entry points to major features.
CourseSelectionFragment	Handles course-selection flow.
YearSelectionFragment	Allows selection of academic year.
TermSelectionFragment	Allows selection of academic term.
ExamAndStuffActivity	Provides examination-related information.
LectureNotesActivity	Provides access to lecture notes.
RoomsAndLabsActivity	Displays room and laboratory information.
RoomNavigateActivity	Supports room navigation.

Developer-Oriented Application Flow

1. Application launch → MainActivity.

- MainActivity initializes the main navigation structure and opens the home/dashboard screen.
- HomeFragment provides entry points to the implemented modules.
- When Course Selection is opened, the user moves through Year Selection, Term Selection, and Course Selection.
- The selected academic information is then used to display the corresponding course information.
- Exam, Lecture Notes, and Rooms & Labs are opened through their respective activities and fragments.
- Room navigation is handled through RoomNavigateActivity when navigation is available.

General User Flow

Course

- Open the app.
- Open Course Selection.
- Choose your year.
- Choose your term.
- Choose a course.
- View the available course information.

Examination

- Open the app.
- Open the Exam section.
- Select the relevant examination information.
- Open and view the details.

Lecture Notes

- Open the app.
- Open Lecture Notes.
- Select the relevant academic/course information.

- Open and read the available note.

Rooms and Labs

- Open the app.
- Open Rooms & Labs.
- Select the required room or laboratory.
- View the information.
- Use navigation when available.

Technology and Implementation Notes

- Platform: Android.
- Architecture style: Activity and Fragment based Android application.
- UI implementation includes generated View/Data Binding classes.
- The APK contains AndroidX-related components.
- Kotlin coroutine-related components are present in the application package.
- Exact backend, API, and database behavior should be confirmed from the source repository and runtime configuration rather than inferred only from the APK.

Known Issue — Course List Year Filtering

- Current behavior: selecting 1st, 2nd, 3rd, or 4th year can result in the same course list being displayed.
- Expected behavior: each selected year should display only the relevant courses.
- Impact: students may see incorrect academic information.
- Recommended investigation: verify that the selected year is captured, passed to the course-list logic, and used to filter the course dataset.
- Testing requirement: repeat the test for all four academic years after the fix to confirm that the correct list appears for each year.

Limitations

- Some planned features are not implemented yet.
- Updates and Settings are not currently part of the implemented feature scope.
- The course-year filtering issue requires correction and regression testing.
- The exact server-side or database architecture cannot be fully established from the APK alone.
- Feature availability may depend on the data included in the current build.

Future Improvements

- Implement Updates.
- Implement Settings.
- Improve course filtering and validation.
- Add stronger error handling and empty-state messages.
- Add automated tests for year, term, and course selection.

- Improve documentation of APIs, data sources, and database structure.
- Add continuous integration and release automation to the GitHub workflow.

Testing and QA

- Test every implemented feature on supported Android versions.
- Test navigation between activities and fragments.
- Verify year and term selections independently.
- Verify that course results match the selected year and term.
- Verify lecture-note and examination content.
- Verify room/lab information and navigation behavior.
- Record reproducible bugs with steps, expected result, actual result, severity, and evidence.

Release and Documentation Status

This document is intended for the KUET Kinship GitHub repository as a README-oriented software documentation reference. It describes the currently implemented scope and separates developer-oriented application flow from the general user flow.

Contribution Guidelines

- Create a separate branch for each feature or bug fix.
- Use clear commit messages.
- Keep changes focused and reviewable.
- Test changes before creating a pull request.
- For bugs, include clear reproduction steps and expected versus actual behavior.
- Update the documentation when a feature or architecture change is introduced.

Conclusion

KUET Kinship provides a focused Android solution for accessing academic and campus information. The current scope includes Course Selection, Examination, Lecture Notes, and Rooms & Labs. The documentation is structured for GitHub/README use and clearly distinguishes implemented features from future work.